FUNDAMENTAL EQUATIONS OF E&M

Maxwell's Equations

Differential form	Integral form	In Matter
$\boldsymbol{\nabla}\cdot\boldsymbol{E}=\frac{\rho}{\epsilon_0}$	$\oint_S \boldsymbol{E}\cdot d\boldsymbol{a}=\frac{Q_{\text{encl}}}{\epsilon_0}$	$\boldsymbol{\nabla}\cdot\boldsymbol{D}=\rho_f$
$\boldsymbol{\nabla}\times\boldsymbol{E}=-\frac{\partial\boldsymbol{B}}{\partial t}$	$\oint_C \boldsymbol{E}\cdot d\boldsymbol{\ell}=-\frac{d\Phi_B}{dt}$	$\boldsymbol{\nabla}\times\boldsymbol{E}=-\frac{\partial\boldsymbol{B}}{\partial t}$
$\boldsymbol{\nabla}\cdot\boldsymbol{B}=0$	$\oint_S \boldsymbol{B}\cdot d\boldsymbol{a}=0$	$\boldsymbol{\nabla}\cdot\boldsymbol{B}=0$
$\boldsymbol{\nabla}\times\boldsymbol{B}=\mu_0\boldsymbol{J}+\mu_0\epsilon_0\frac{\partial\boldsymbol{E}}{\partial t}$	$\oint_C \boldsymbol{B}\cdot d\boldsymbol{\ell}=\mu_0\Phi_J+\mu_0\epsilon_0\frac{d\Phi_E}{dt}$	$\boldsymbol{\nabla}\times\boldsymbol{H}=\boldsymbol{J}_f+\frac{\partial\boldsymbol{D}}{\partial t}$

Potentials

$$\boldsymbol{E}=-\boldsymbol{\nabla}V-\frac{\partial\boldsymbol{A}}{\partial t} \qquad \boldsymbol{B}=\boldsymbol{\nabla}\times\boldsymbol{A} \qquad \boldsymbol{H}=-\boldsymbol{\nabla}\phi_m$$

Auxiliary Fields

General Equations	Linear Media	
$\boldsymbol{D}=\epsilon_0\boldsymbol{E}+\boldsymbol{P}$	$\boldsymbol{P}=\epsilon_0\chi_e\boldsymbol{E}$	$\boldsymbol{D}=\epsilon\boldsymbol{E}=\kappa\epsilon_0\boldsymbol{E}$
$\boldsymbol{H}=\frac{1}{\mu_0}\boldsymbol{B}-\boldsymbol{M}$	$\boldsymbol{M}=\chi_m\boldsymbol{H}$	$\boldsymbol{H}=\frac{1}{\mu}\boldsymbol{B}=\frac{1}{\kappa_m\mu_0}\boldsymbol{B}$
$\boldsymbol{F}=q(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B})$	$\boldsymbol{F}=q(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B})$	
$u=\frac{1}{2}\rho V+\frac{1}{2}\boldsymbol{J}\cdot\boldsymbol{A}$		
$u=\frac{1}{2}\epsilon_0E^2+\frac{1}{2\mu_0}B^2$	$u=\frac{1}{2}\boldsymbol{D}\cdot\boldsymbol{E}+\frac{1}{2}\boldsymbol{H}\cdot\boldsymbol{B}$	
$\boldsymbol{S}=\frac{1}{\mu_0}\boldsymbol{E}\times\boldsymbol{B}$	$\boldsymbol{S}=\boldsymbol{E}\times\boldsymbol{H}$	

Current

$$\boldsymbol{\nabla}\cdot\boldsymbol{J}=-\frac{\partial\rho}{\partial t} \qquad \boldsymbol{J}=\sigma\boldsymbol{E} \qquad \text{Power}=\int \boldsymbol{J}\cdot\boldsymbol{E}\,d^3r$$

BOUNDARY CONDITIONS

($\hat{\boldsymbol{n}}$ points 1 → 2)

$$V^2-V^1=0$$

$$E_n^2-E_n^1=\frac{\sigma}{\epsilon_0} \qquad D_n^2-D_n^1=\sigma_f$$

$$\boldsymbol{E}_t^2-\boldsymbol{E}_t^1=0 \qquad \boldsymbol{D}_t^2-\boldsymbol{D}_t^1=\boldsymbol{P}_t^2-\boldsymbol{P}_t^1$$

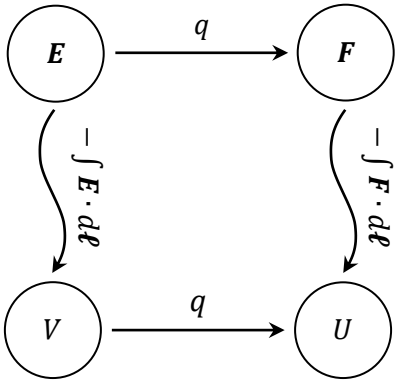
$$\phi_m^2-\phi_m^1=0 \text{ if } \boldsymbol{K}_f=0$$

$$A_n^2-A_n^1=0 \quad \text{if } \boldsymbol{\nabla}\cdot\boldsymbol{A}=0$$

$$\boldsymbol{A}_t^2-\boldsymbol{A}_t^1=0$$

$$B_n^2-B_n^1=0 \qquad H_n^2-H_n^1=-(M_n^2-M_n^1)$$

$$\boldsymbol{B}_t^2-\boldsymbol{B}_t^1=\mu_0\boldsymbol{K}\times\hat{\boldsymbol{n}} \qquad \boldsymbol{H}_t^2-\boldsymbol{H}_t^1=\boldsymbol{K}_f\times\hat{\boldsymbol{n}}$$



In this space, I use a modified version of the ρ, V, $\boldsymbol{E}$ triangle from *Introduction to Electrodynamics*, 4th ed., by David J. Griffiths, Pearson, 2013, Fig 2.35, pg 88

In the space, I use a modified version of the $\boldsymbol{J}$, $\boldsymbol{A}$, $\boldsymbol{B}$ triangle from *Introduction to Electrodynamics*, 4th ed., by David J. Griffiths, Pearson, 2013, Fig 5.48, pg 250

Calculation Strategy Summary

Method		Electric Field $\boldsymbol{E}$	Electric Potential V
C.	Connection	$\boldsymbol{E} = -\boldsymbol{\nabla} V$	$\Delta V = -\int \boldsymbol{E}\cdot d\boldsymbol{\ell}$
S.	Superposition	any combination of below	any combination of below
1.	Direct	$\boldsymbol{E} = kq/r^2\,\hat{\boldsymbol{r}}$	$V = kq/r$
D 2.	Gauss's law	choose surface →	C.
3.	Meth. of Images	choose images → 1. Direct	choose images → 1. Direct
D 4.	Sep. of Var.	C. ←	sep. of var.
5.	Multipole Exp.	C. ←	calculate multipoles

Method		Magnetic Field $\boldsymbol{B}$	Vector Potential $\boldsymbol{A}$
C.	Connection	$\boldsymbol{B} = \boldsymbol{\nabla}\times\boldsymbol{A}$	Ampere's law for $\boldsymbol{A}$
S.	Superposition	any combination of below	any combination of below
1.	Direct	$\boldsymbol{B} = \frac{\mu_0}{4\pi}\int \frac{\boldsymbol{J}\times\hat{\boldsymbol{r}}}{r^2}d^3r'$	$\boldsymbol{A} = \frac{\mu_0}{4\pi}\int \frac{\boldsymbol{J}}{r}d^3r'$
H 2.	Ampere's law	choose loop →	C.
3.	~~Meth. of Images~~		
H 4.	Scalar Pot. ϕ_m	C. $\boldsymbol{B} = -\mu\,\boldsymbol{\nabla}\phi_m$ ←	sep. of var. (for ϕ_m)
5.	Multipole Exp.	C. ←	calculate multipoles

CALCULATION METHODS (CONT.)

Separation of Variables – Spherical

$$V(r,\theta) = \sum_{\ell=0}^{\infty} \left(A_\ell r^\ell + B_\ell \frac{1}{r^{\ell+1}} \right) P_\ell(\cos\theta)$$

Fourier's trick

If $f(\theta) = \sum_{\ell=0}^{\infty} C_\ell P_\ell(\cos\theta)$

then $C_\ell = \frac{2\ell+1}{2} \int_0^\pi f(\theta) P_\ell(\cos\theta) \sin\theta \, d\theta$

Legendre Polynomials

$P_0(u) = 1$ $\qquad$ $P_2(u) = \frac{3}{2}u^2 - \frac{1}{2}$

$P_1(u) = u$ $\qquad$ $P_3(u) = \frac{5}{2}u^3 - \frac{3}{2}u$

Monopoles

$$Q = \int \rho(\boldsymbol{r}') \, d^3r' \qquad V(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

Dipoles

$$\boldsymbol{p} = \int \boldsymbol{r}' \rho(\boldsymbol{r}') \, d^3r' \qquad \boldsymbol{m} = \frac{1}{2} \int \boldsymbol{r}' \times \boldsymbol{J}(\boldsymbol{r}') \, d^3r'$$

$$V(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0} \frac{\boldsymbol{p} \cdot \hat{\boldsymbol{r}}}{r^2} \qquad \boldsymbol{A}(\boldsymbol{r}) = \frac{\mu_0}{4\pi} \frac{\boldsymbol{m} \times \hat{\boldsymbol{r}}}{r^2}$$

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0} \frac{3(\boldsymbol{p} \cdot \hat{\boldsymbol{r}})\hat{\boldsymbol{r}} - \boldsymbol{p}}{r^3} \qquad \boldsymbol{B}(\boldsymbol{r}) = \frac{\mu_0}{4\pi} \frac{3(\boldsymbol{m} \cdot \hat{\boldsymbol{r}})\hat{\boldsymbol{r}} - \boldsymbol{m}}{r^3}$$

Quadrupoles

$$Q_{ij} = \int \frac{3r_i' r_j' - r'^2 \delta_{ij}}{2} \rho(\boldsymbol{r}') \, d^3r' \qquad V(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0} \frac{\sum_{ij} Q_{ij} \hat{r}_i \hat{r}_j}{r^3}$$

Method of Images

$$q' = -qR/z_0 \qquad z_0' = R^2/z_0$$

Bound Charge and Current

$$\sigma_b = \boldsymbol{P} \cdot \hat{\boldsymbol{n}} \qquad \boldsymbol{K}_b = \boldsymbol{M} \times \hat{\boldsymbol{n}}$$

$$\rho_b = -\boldsymbol{\nabla} \cdot \boldsymbol{P} \qquad \boldsymbol{J}_b = \boldsymbol{\nabla} \times \boldsymbol{M}$$

COORDINATES

$x = \rho\cos\phi$ $\quad\quad$ $\hat{\boldsymbol{x}} = \cos\phi\,\hat{\boldsymbol{\rho}} - \sin\phi\,\hat{\boldsymbol{\phi}}$

$y = \rho\sin\phi$ $\quad\quad$ $\hat{\boldsymbol{y}} = \sin\phi\,\hat{\boldsymbol{\rho}} + \cos\phi\,\hat{\boldsymbol{\phi}}$

$x = r\sin\theta\cos\phi$ $\quad\quad$ $\hat{\boldsymbol{x}} = \sin\theta\cos\phi\,\hat{\boldsymbol{r}} + \cos\theta\cos\phi\,\hat{\boldsymbol{\theta}} - \sin\phi\,\hat{\boldsymbol{\phi}}$

$y = r\sin\theta\sin\phi$ $\quad\quad$ $\hat{\boldsymbol{y}} = \sin\theta\sin\phi\,\hat{\boldsymbol{r}} + \cos\theta\sin\phi\,\hat{\boldsymbol{\theta}} + \cos\phi\,\hat{\boldsymbol{\phi}}$

$z = r\cos\theta$ $\quad\quad$ $\hat{\boldsymbol{z}} = \cos\theta\,\hat{\boldsymbol{r}} - \sin\theta\,\hat{\boldsymbol{\theta}}$

VECTOR IDENTITIES

Triple Products

(1) $\boldsymbol{A}\cdot(\boldsymbol{B}\times\boldsymbol{C}) = \boldsymbol{B}\cdot(\boldsymbol{C}\times\boldsymbol{A}) = \boldsymbol{C}\cdot(\boldsymbol{A}\times\boldsymbol{B})$

(2) $\boldsymbol{A}\times(\boldsymbol{B}\times\boldsymbol{C}) = \boldsymbol{B}(\boldsymbol{A}\cdot\boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A}\cdot\boldsymbol{B})$

Product Rules

(3) $\boldsymbol{\nabla}(fg) = f(\boldsymbol{\nabla}g) + g(\boldsymbol{\nabla}f)$

(4) $\boldsymbol{\nabla}(\boldsymbol{A}\cdot\boldsymbol{B}) = \boldsymbol{A}\times(\boldsymbol{\nabla}\times\boldsymbol{B}) + \boldsymbol{B}\times(\boldsymbol{\nabla}\times\boldsymbol{A}) + (\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B} + (\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A}$

(5) $\boldsymbol{\nabla}\cdot(f\boldsymbol{A}) = f(\boldsymbol{\nabla}\cdot\boldsymbol{A}) + \boldsymbol{A}\cdot(\boldsymbol{\nabla}f)$

(6) $\boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A}) - \boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B})$

(7) $\boldsymbol{\nabla}\times(f\boldsymbol{A}) = f(\boldsymbol{\nabla}\times\boldsymbol{A}) - \boldsymbol{A}\times(\boldsymbol{\nabla}f)$

(8) $\boldsymbol{\nabla}\times(\boldsymbol{A}\times\boldsymbol{B}) = (\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A} - (\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B} + \boldsymbol{A}(\boldsymbol{\nabla}\cdot\boldsymbol{B}) - \boldsymbol{B}(\boldsymbol{\nabla}\cdot\boldsymbol{A})$

Second Derivatives

(9) $\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A}) = 0$

(10) $\boldsymbol{\nabla}\times(\boldsymbol{\nabla}f) = 0$

(11) $\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{A}) = \boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) - \nabla^2\boldsymbol{A}$

Integral Theorems

$$\int_V \boldsymbol{\nabla}\cdot\boldsymbol{F}\,d^3r = \oint_{\partial V}\boldsymbol{F}\cdot d\boldsymbol{a}$$

$$\int_S (\boldsymbol{\nabla}\times\boldsymbol{F})\cdot d\boldsymbol{a} = \oint_{\partial S}\boldsymbol{F}\cdot d\boldsymbol{\ell}$$

DIFFERENTIAL OPERATORS

$$d\boldsymbol{\ell} = \hat{\boldsymbol{x}}\,dx + \hat{\boldsymbol{y}}\,dy + \hat{\boldsymbol{z}}\,dz$$
$$= \hat{\boldsymbol{\rho}}\,d\rho + \hat{\boldsymbol{\phi}}\,\rho d\phi + \hat{\boldsymbol{z}}\,dz$$
$$= \hat{\boldsymbol{r}}\,dr + \hat{\boldsymbol{\theta}}\,rd\theta + \hat{\boldsymbol{\phi}}\,r\sin\theta\,d\phi$$

$$\boldsymbol{\nabla}f = \hat{\boldsymbol{x}}\frac{\partial f}{\partial x} + \hat{\boldsymbol{y}}\frac{\partial f}{\partial y} + \hat{\boldsymbol{z}}\frac{\partial f}{\partial z}$$
$$= \hat{\boldsymbol{\rho}}\frac{\partial f}{\partial \rho} + \hat{\boldsymbol{\phi}}\frac{1}{\rho}\frac{\partial f}{\partial \phi} + \hat{\boldsymbol{z}}\frac{\partial f}{\partial z}$$
$$= \hat{\boldsymbol{r}}\frac{\partial f}{\partial r} + \hat{\boldsymbol{\theta}}\frac{1}{r}\frac{\partial f}{\partial \theta} + \hat{\boldsymbol{\phi}}\frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$
$$= \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho A_\rho) + \frac{1}{\rho}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$
$$= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\,A_\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial \phi}$$

$$\boldsymbol{\nabla}\times\boldsymbol{A} = \hat{\boldsymbol{x}}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{\boldsymbol{y}}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{\boldsymbol{z}}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$
$$= \hat{\boldsymbol{\rho}}\left(\frac{1}{\rho}\frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z}\right) + \hat{\boldsymbol{\phi}}\left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho}\right) + \hat{\boldsymbol{z}}\frac{1}{\rho}\left(\frac{\partial}{\partial \rho}(\rho A_\phi) - \frac{\partial A_\rho}{\partial \phi}\right)$$
$$= \hat{\boldsymbol{r}}\frac{1}{r\sin\theta}\left(\frac{\partial}{\partial \theta}(\sin\theta\,A_\phi) - \frac{\partial A_\theta}{\partial \phi}\right) + \hat{\boldsymbol{\theta}}\left(\frac{1}{r\sin\theta}\frac{\partial A_r}{\partial \phi} - \frac{1}{r}\frac{\partial}{\partial r}(rA_\phi)\right) + \hat{\boldsymbol{\phi}}\frac{1}{r}\left(\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial \theta}\right)$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$
$$= \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\frac{\partial f}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}$$
$$= \frac{1}{r}\frac{\partial^2}{\partial r^2}(rf) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial \phi^2}$$